FSD
Semester 1, 2026

Lectorial Week 01

Introduction to React

How React works?

Typescript

React components

RMIT UNIVERSITY

COMPUTING TECHNOLOGIES

CRICOS provider number: 00122A | RTO Code: 3046

Womin djeka

RMIT University acknowledges the people of the Woi wurrung and Boon wurrung language groups of the eastern Kulin Nation on whose unceded lands we conduct the business of the University.

RMIT respectfully acknowledges their Ancestors and Elders, past and present.

RMIT also acknowledges the Traditional Custodians and their Ancestors of the lands and waterways across Australia where we conduct our business.

Segment 1: Administrivia

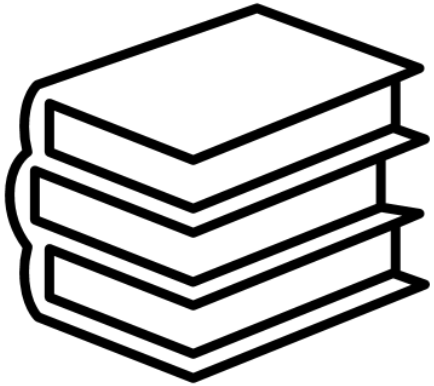
Lectorial

Lab

Assignments

Expectations

Introduction



Staff

- **Course Coordinator and Lecturer:**
  - **Dipto Pratyaksa**
    - **dipto.pratyaksa@rmit.edu.au**
    - **Normal work hours**: Monday-Friday 8:30-4:30pm + late afternoon classes. I don't open emails outside of these hours.
    - **Consultation hours:** Tuesday-Thursday 11:00am-12:30pm.
    - Book time with Dipto Pratyaksa [https://outlook.office.com/bookwithme/user/4631ea1b27484e16bbc88d3428e83ca4@rmit.edu.au?anonymous&ismsaljsauthenabled&ep=pcard]

- For a detailed list of staff members teaching (tutors) in this course, please look at
  - Course Canvas -> Modules-> Teaching Team *page*

Why was this course created?

- Feedback from industry who hired RMIT graduates
- Feedback from RMIT graduates
- A gap in the student skill- *full stack development*
- Teach React or Angular so that graduates can pick up other frameworks as well (such as Vue.js, etc.)
- There was always a demand from students to bring back the course
- React programming is a sought-after skill

General tone of this course

- Highly practical and *generally* intense
- Easy to semi-easy to get Pass
- Semi-easy to moderate effort to get Credit
- Need to work hard, research and find answers to get Distinction and High Distinction
- Two assignments worth 70% - related to each other
- <u>Work in a group of 2 (a pair)</u>
- Use of GitHub Classroom for all assignments and weekly pracs
- Individual Final assessment (time allowed: 2 hours) at the end of semester worth 30%

Important Advice about A1 and A2

- Assignment 1:
  - based on week(s) 1 till 5
  - worth 25%
- Assignment 2: *based on Assignment 1*
  - based on week(s) 5 till 11
  - worth 45%
- DI and HD parts of assignments will require you to research and find out answers
- We will help but you will drive the work
- This has been done on purpose to make you work-ready and prepare for industry

Pre-reading, Lectorial, Lab

- Pre-reading: always read/watch the recommended pre-reading material before attending the lectorial
- Lectorial: will cover the concepts with code examples along with interactive discussion
- Lab/Prac: will cover application of concepts covered during lectorial
- Lab/Prac start week 1! Important to attend week 1 classes to cover the fundamentals and tools administration.
- Week 2 will be intense, so you need the right gadgets, software tools and **mindset** to do the job.

How do I get good grade in this course?

- Attend all lectorials and labs. Listen well.
- Practise code each week
- Start working on assessments on time
  - Last minute work can lead to undesirable outcomes
- Try to understand what you code
  - Copying code from various sources without understanding can lead to disastrous outcomes
- Ask questions
  - Keeping quiet and leaving things till the end can lead to unwanted scenarios

What is this course about?

✓	✗
React framework	Web design
Coding every aspect of the web app	Using drag and drop tools
Learning a web framework	Not attending any class or read course material
Acquiring a good knowledge of a web framework	Use of frameworks that are not covered in this course
Learning by doing things yourself	Reading concepts only and not coding
Asking questions and attending consults	Keeping quiet and not participating

Few reasons for failing the course among previous cohorts

- *I did not attend any session, I assumed I knew enough to pass* ✗
- *I had coded in React in one of my earlier projects, so I thought how hard could it be* ✗
- *I assumed that I could complete the assessment in this way* ✗
- *Writing code in a framework other than React* ✗
- *I was not sure if I had to follow the assignment specifications exactly* ✗
- *I found React hard and I just gave up* ✗

Prescribed textbook?

- No one book has all the topics
- Web development is a fast-moving field
- It is not possible to buy multiple textbooks
- You will be given handy references of free online references as supplementary material

Before we start

- basic html tutorial [https://www.w3schools.com/html/]:
  - read sections *HTML Home to Html Lists*
- basic CSS tutorial [https://www.w3schools.com/css/default.asp]:
  - read sections *CSS home to CSS Backgrounds, CSS text to CSS Lists*
- basic JS tutorial [https://www.w3schools.com/js/DEFAULT.asp]
- jQuery (helpful to know event-driven programming)
- have a basic knowledge of SQL

Prerequisites for COSC2758

Undergrad students enrolled in Full Stack Development must have passed at least one of the following courses:

- COSC2803 - Java Programming Studio (Course ID 054081)
- COSC2413 / COSC2430 / COSC2453 / COSC2690 - Web Programming (Course ID 039990)
- EEET2482 - Software Engineering Design (Course ID 038296)
- COSC3046/COSC3060/COSC3061 - Web Programming Studio (Course ID 054990)

Prerequisites for COSC2938

Postgrad students enrolled in Further Web Programming must have passed at least one of the following courses:

- COSC1295 Advanced Programming (Course ID 004316)
- COSC3015/COSC2820 Advanced Programming for Data Science (Course ID 054137)

Please talk to RMIT STEM Student Lifecycle / your course program manager if you have not passed the prerequisite.

Segment 2

Introduction- *why?*

React- *what?*

React- *how?*

The world of components

Why a web framework?

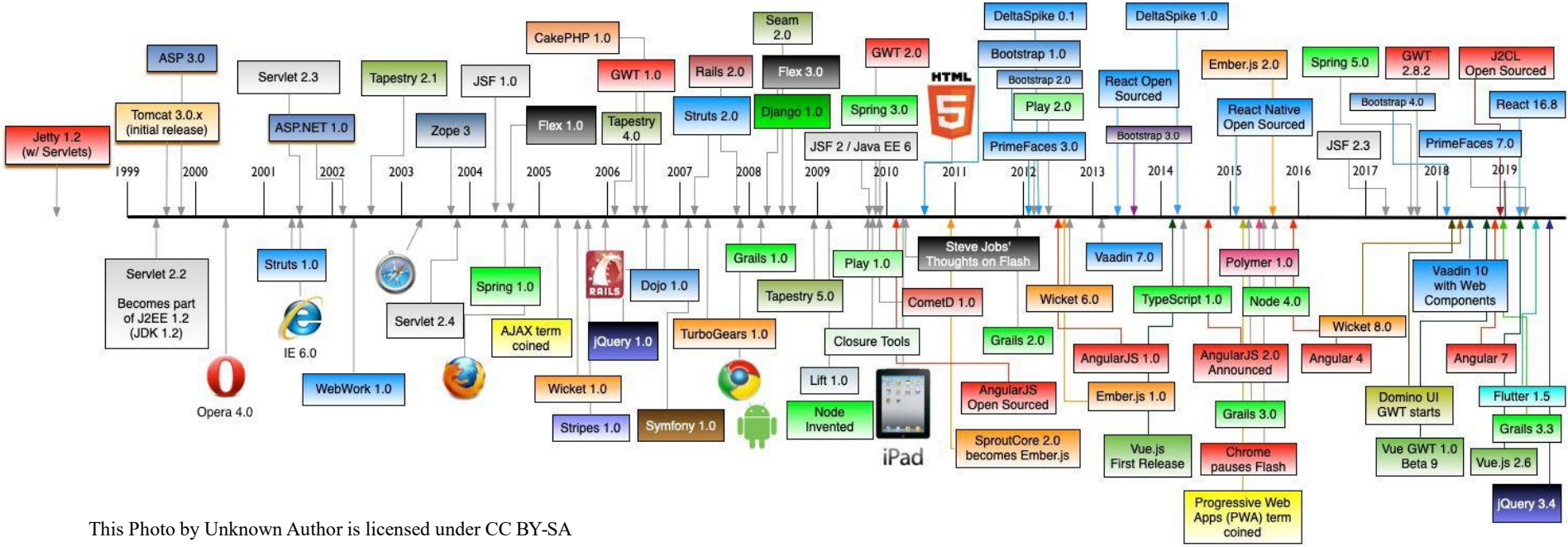
- Surely, one can design a website with just *HTML5, CSS* and *JavaScript*
- Then why learn something new? Besides, *a framework sounds complicated!*
  - Real life/ industry web application projects are complex
  - There is a need for automating repetitive tasks
  - Web is growing and so are the needs, technologies
  - Multiple tiers/stacks are involved

Why a web framework?

- Frameworks simplify the previously mentioned scenarios
- Abstracts unnecessary details from the user
- It provides templates for the user
- It handles all the logistical aspects of the application development
- In essence, it leads to
  - Accelerated development &
  - Efficient debugging

Many web frameworks and libraries

- ASP.NET, JEE (Java Enterprise Ed.)
- React, Angular, Vue, Django, Ruby on Rail, Laravel, etc.
- Flask, Express.js, jQuery, etc.



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Skills for the CSIT industry

- Technical
  - Technology, frameworks, languages, *etc.*
- Graduate attributes
  - EQ, group dynamics, humble disposition, collaboration skills, initiative, creativity, innovation etc.
- *Complete turnoff(s) / incorrect mindset:*
  - *Arrogance / Know-it-all*
  - *Reluctance to learn anything new*
  - *Not open to working alongside others*
  - *Negative attitude*

Why React? Why not any other?

- It is best to learn few frameworks in an efficient manner
- This way you can pick up on others- *this is how industry works!*
- React is a library, great as starting point
- Angular / Vue.js are good frameworks to learn
- <u>We will focus on React + supporting web libraries and frameworks in this course</u>
- You are expected to write all assessments in React and not in any unrelated framework of your choice
- <u>You will get ZERO if you went the non-React way</u>!

Framework vs Library: The Key Distinctions

Aspect	Framework	Library
Control Flow	The framework dictates the overall flow and architecture of the application.	The developer controls the application's flow and calls library functions as needed.
Structure	Imposes a specific architecture, rules, and conventions (e.g., MVC or MTV patterns).	Offers functions and classes without imposing a particular project structure.
Scope	Comprehensive; designed to build entire applications and handle many aspects of development (routing, security, etc.).	Focused; designed for specific, well-defined tasks (e.g., image manipulation, date formatting, data manipulation).
Flexibility	Less flexible; developers must adhere to the framework's guidelines and structure.	More flexible; can be easily integrated into an existing codebase and combined with other tools.
Learning Curve	Generally has a steeper learning curve due to its conventions and integrated tools.	Generally has a lower learning curve, as it focuses on specific, isolated tasks.
Replaceability	Difficult to replace without rewriting a significant portion of the codebase.	Easy to replace with another library that serves a similar purpose.

What is *Full stack development*?

- The development of both front end(*client side*) and back end(*server side*) portions of a web application.
- A true full stack developer is proficient in both frontend and backend languages and frameworks, as well as in server, network and setting, hosting environments.
- Technology stacks:

1. HTML, CSS, Bootstrap, React in Typescript *(frontend)*
2. Node.js & Express.js in Typescript *(backend)*
3. Database: MySQL / MSSQL *(data storage)*

You probably use it every day

- Facebook
- Netflix
- AirBnB
- UberEats
- Dropbox
- PayPal
- Reddit
- Shopify

React is used in 43% of the Top 10,000 websites according to Built With (https://trends.builtwith.com/javascript/React)

Not just websites

- Productivity and Design Web apps: Figma, Canva, Slack, Eagle, Remotion | Make videos programmatically, ScreenStudio
- AI & Development tools: Cursor, Claude Code, Datics.ai, Streamlit, Expo
- Command Line Interpreter: Ink vadimdemedes/ink: 🌈 React for interactive command-line apps
- Some of these web apps are also available as mobile apps developed using React Native Showcase · React Native

React - introduction

- A **frontend JavaScript library** created and maintained by Facebook (now Meta)
- Popular and strong demand in industry:
  - 42-44% of professional developers use React.js
  - Top choice for frontend frameworks:
    - 22 million weekly NPM downloads
    - 200,000+stars on GitHub
  - Job market: represent 50% of front-end developer positions, 847,000+ job postings globally, with **67% year-over-year growth** in demand

React - introduction

- Industry Adoption & Trends:
  - React is heavily adopted in Technology & Software (32%), E-commerce & Retail (18%), Financial Services (15%), and Healthcare (12%).
  - Next.js is now heavily adopted by 67% of React developers, serving as primary framework for SSR (Server-Side Rendering)
  - Mobile Market: React Native powers over 200,000+ apps.
- Key Competitors: Angular, Vue.js, Flutter
- Bright Future Outlook:
  - Market size projected to reach $28.6 billion USD
  - Critical skill for developers.
  - Strong demand for expertise in modern features like Server Components, React 19 and Typescript

React - introduction

- Super flexible: You can integrate it with many libraries and frameworks within or outside React ecosystem
- Helps creating web applications
- You can also learn React *Native* in your free time (building cross platform mobile apps)
- JavaScript (JS) or TypeScript (TS) based- these are not hard to learn
- Learning React will help you learn/pickup other libraries and frameworks

React with Typescript in this course

- We will learn how to create React application with Typescript (TS)
- Many reasons for doing this
  - TS is an object-oriented programming language developed by Microsoft.
  - TS is a superset (the parent) of JS with generic and JS features to overcome the complexities of JS.
  - Whilst slightly higher learning curve than JS, quite easy coming from other typed languages such as Java
  - Additionally helps correct common errors that you might face (not passing enough arguments, wrong data types etc)

Quick intro to Typescript - Typing

TypeScript allows you to specify types for variables, function parameters, and return values.

typescript Copy Edit

```typescript
let message: string = "Hello, TypeScript!";
let age: number = 30;
let isDone: boolean = false;
```

Without type annotations, TypeScript will infer the type.

Functions in Typescript

typescript Copy Edit

```typescript
function add(a: number, b: number): number {
  return a + b;
}
```

- `a` and `b` must be numbers.
- The return type is `number`.

Optional and default parameters:

typescript Copy Edit

```typescript
function greet(name: string, greeting: string = "Hello"): string {
  return `${greeting}, ${name}!`;
}

greet("Alice"); // "Hello, Alice!"
greet("Bob", "Hi"); // "Hi, Bob!"
```

Interfaces in Typescript

Define object shapes using **interfaces**.

typescript Copy Edit

```typescript
interface User {
  name: string;
  age: number;
}

const user: User = { name: "Alice", age: 25 };
```

Interfaces support **optional** properties (`?`) and **readonly** properties.

typescript Copy Edit

```typescript
interface Car {
  readonly model: string;
  year?: number; // Optional property
}
```

Arrays in Typescript

typescript Copy Edit

```typescript
let numbers: number[] = [1, 2, 3];
let names: string[] = ["Alice", "Bob"];

let person: [string, number] = ["Alice", 25]; // Tuple (fixed types & order)
```

React – why?

- Single-page applications (SPA) have become increasingly popular with first-generation SPA frameworks like Angular (by Google), etc.
- Although you can create these with vanilla HTML, CSS, JS, but the process becomes cumbersome
- React helps you create SPAs?
- Why SPAs?
  - In the past, websites and web applications were rendered from the server.
  - Modern frameworks shifted the focus from the server to the client
  - Examples: Gmail, Google Maps, Airbnb, Netflix, PayPal, etc.

Lectorial Exercise

- Why use SPA as compared to traditional websites?
- Provide few examples of popular SPA websites.
- What are some disadvantages of SPA websites?

React- setup

- You will need a laptop/machine to complete this elective course - RMIT labs/myDesktop does not have a React setup
- Download Node.js- latest version – *free to install*
  - [https://nodejs.org/en/download/]
  - If already installed on your machine, update it to the latest version.
- Download Visual Studio code- IDE used for this course – *free to install*
  - [https://code.visualstudio.com/download]

Lectorial Exercise

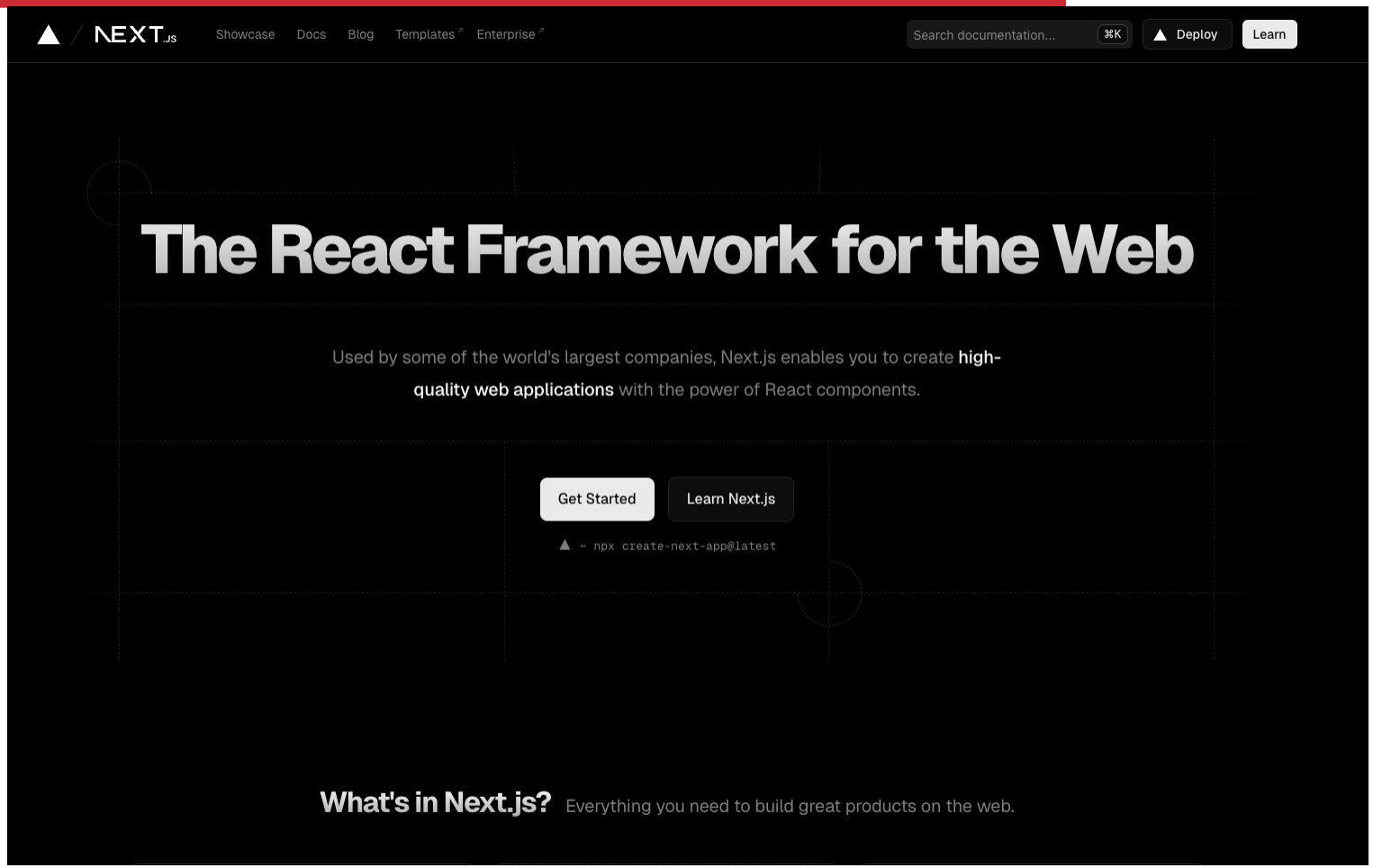
- What is Node.js?
- What is it used for?
- Why use it for React?

Node & NPM

- Installation of Node.js will also setup NPM (Node package Manager)
- You will need both Node and NPM to manage libraries (node packages) that you will need along the way.
- Post installation, check that you have the latest version (Node 22) by running these commands-command prompt (windows), Terminal (Mac, Linux):

```
node --version
*vXX.YY.ZZ
npm --version
*vXX.YY.ZZ
```

NextJS


NEXT.JS
Showcase
Docs
Blog
Templates
Enterprise
Search documentation...
⌘K
Deploy
Learn
The React Framework for the Web
Used by some of the world's largest companies, Next.js enables you to create high-quality web applications with the power of React components.
Get Started
Learn Next.js
~ npx create-next-app@latest
What's in Next.js?
Everything you need to build great products on the web.

Setting up a React project

- Run this command on command prompt/terminal:
  - `npx create-next-app@latest`
  - This will take sometime depending upon various factors- so do not give up!
  - Select all of the defaults (ie keep hitting enter, just make sure App Router is set to No)
- Once created
  - `cd my-app`
  - `npm run dev`
- Example01

Example 1- *project structure*

```
my-next-app/
|— public/            # Static assets (images, fonts, etc.)
|— src/               # (Optional) Organizes app code
|    ├— pages/        # (For Pages Router) Route-based components
|    ├— components/   # Reusable UI components
|    ├— styles/       # Global/local styles (CSS, SCSS, Tailwind, etc.)
|    ├— lib/          # Utilities, API clients, or helpers
|    ├— hooks/        # Custom React hooks
|    ├— context/      # Global state management (e.g., React Context)
|    ├— middleware/   # Custom middleware (if needed)
|    ├— api/          # Serverless functions (in pages/api for Pages Router)
|— .next/             # Build output (generated at runtime)
|— node_modules/      # Installed dependencies
|— .env.local         # Environment variables
|— next.config.js     # Next.js configuration file
|— package.json       # Project dependencies and scripts
|— tsconfig.json      # TypeScript configuration (if using TS)
|— README.md          # Project documentation
```

Component *eh!*

- Every React application is built on the foundation of React components.
- Your first React component which is located in the src/pages/index.tsx

```
1   export default function Home() {
2     return <div>Hello World</div>;
3   }
```

Default template of a component

- First, this React component, called the App component, is just a JavaScript / Typescript function.
- Second, the App component doesn't receive any parameters in its function signature yet.
- And third, the App component returns code that resembles HTML-
  - this new syntax is called JSX (or TSX for Typescript), allows you to combine JavaScript and HTML for displaying highly dynamic and interactive content in a browser.

☺

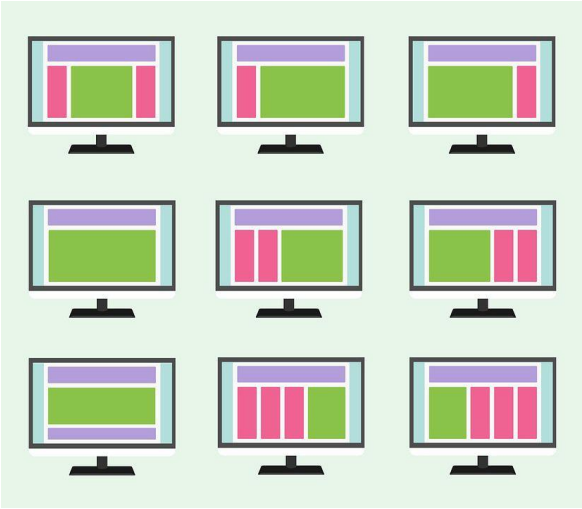
What is the meaning of this?
HTML
React
JS

React JSX

- Until now, we only returned HTML from the App component.
- However, it can also be mixed with JavaScript (JS).
- JSX helps you to combine HTML and JS
- Without any extra templating syntax you will be able to use JavaScript in HTML.
- Every data structure, from JavaScript primitive to a JavaScript function, can be displayed within HTML with the help of JSX
- Let us now make our component a bit smarter
- Example 2

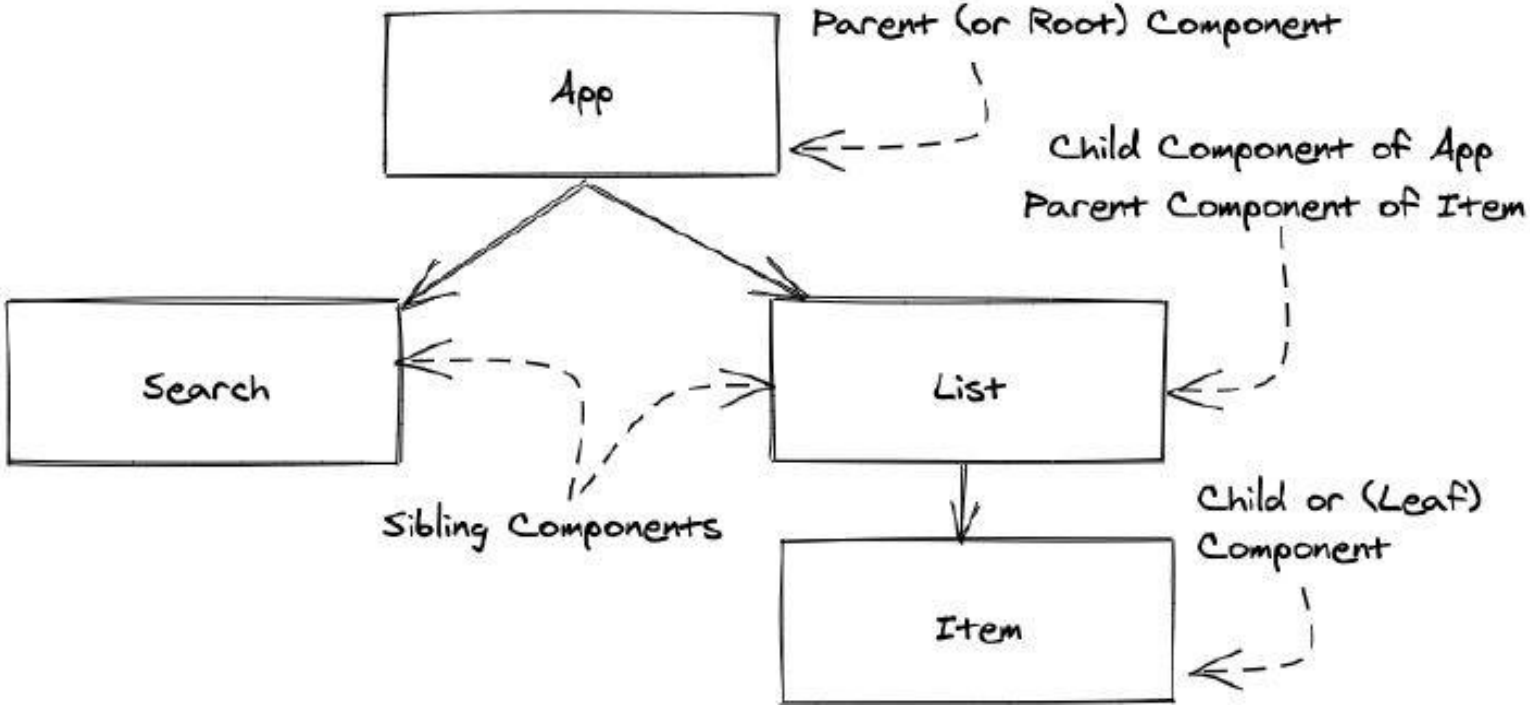
Component or Components

- So far we've only been using the App component.
- This will never be the case in case of a website which is more complex than a single component
- So instead of making one component larger and more complex, we'll split one component into multiple components eventually



Component or Components

- We will now look at more involved example where we will develop multiple components.



Component or Components

- React applications have component hierarchies (also called component trees).
- There is usually one uppermost entry point component (e.g. App) that spans a tree of components below it.
- The App is the parent component of the List and Search, so the List and Search are child components of the App component and sibling components to each other.
- Example 3

Can I use class components for this course?

- In this course we will only learn how to write new React i.e. *functional* components
- You are NOT allowed OR advised to write class components
  - You will get a zero in assessments for using class components
- Only legacy projects use class components
- So remember
  - Class components = ZERO

Lectorial Exercise

- What is DOM?
- Where have you used it?

Handling events

- Event handling makes it possible for your users to interact with the React app
- As per React documentation- handling events with React elements is similar to handling events on DOM elements except
  - React events are named using camelCase, rather than lowercase.
  - With JSX you pass a function as the event handler, rather than a string
- Assuming you're familiar with how events work in standard HTML and JavaScript, it should be easy for you to learn how to handle events in React.

Handling events: functional component

- This is not very different to a class component case
- Example04

Hooks

State in functional components

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Lectorial Exercise

- **What is a *state* in a web application?**
- **Why is it needed?**

State in functional components: *hooks*

- They let you use state and other React features without writing a class.
- They simplify a lot of code writing scenarios in React
- In-built hooks in React
  - https://reactjs.org/docs/hooks-reference.html

- Basic Hooks
  - `useState`
  - `useEffect`
  - `useContext`
- Additional Hooks
  - `useReducer`
  - `useCallback`
  - `useMemo`
  - `useRef`
  - `useImperativeHandle`
  - `useLayoutEffect`
  - `useDebugValue`

useState()

- It returns a stateful value

  const [state, setState] = useState(initialState);
- The setState function is used to update the state

  setState(newState);
- Time to look at an example
- Example05

- We will keep revisiting hooks in subsequent weeks- there is a lot to cover

useState() *nuts & bolts*

1. Call useState() hook to enable state in a functional component.

2. The first argument of the useState(initialValue) is the state's initial value.

3. [state, setState] = useState(initialValue) returns an array of 2 items: the state value and a state updater function.

useState() *nuts & bolts*

4. Invoking the state updater function setState(newState) with the new value updates the state.
    - Alternatively, you can invoke the state updater with a callback setState(prev => next), which returns the new state based on previous.

5. After the state updater is called, React makes sure to re- render the component so that the new state becomes actual.

So what is a *hook*?

- **What is a Hook?**
  - A *Hook* is a special function that lets you “hook into” React features. For example, *useState* is a Hook that lets you add React state to function components.
- **When would I use a Hook?**
  - If you write a function component and realize you need to add some state to it, previously you had to convert it to a class. Now you can use a Hook inside the existing function component.

Reference:

https://reactjs.org/docs/hooks-reference.html

References

- The Road to React (2021 edition), by Robin Weiruch; Leanpub
- The Road to React: The React.js 19 with Hooks in JavaScript (2025 edition), by Robin Weiruch; Leanpub
- “React developers market share” Google AI Mode [https://share.google/aimode/fSzLJ4hWIwGdRgGSM]
- TypeScript vs JavaScript: Which One Is Better to Choose? by Milit Panchasara (Radix)

Next week

- More on React components
- Components interacting - part 1
- Incorporating data
- Styling in React &
- Assignment 1 discussion